

gLike Updates

Summary

- Increased `kappa` from 10,000 to 50,000
- Decreased iteration count from 20 to 10 for `gLike` optimizer (limitations of feasibility with increased `kappa`)
- Issue: CMA-ES struggles with certain boundary conditions.
 - Previously, boundary conditions in CMA-ES are separate from boundary conditions as defined in `gLike` .
 - I more robustly coded the requirements for boundary conditions under CMA-ES; however, we now run into an issue where, even when expanding the parameter solution space (via `kappa`), it is now possible to now have any viable solution.
 - Example 1, completely ignoring migration:

```
2026-09-23 11:03:55,096 - INFO - Starting glike optimization.
```

```
-118373.61748614724
```

```
INVALID PARAMETERS:
```

```
gr_chb out of bounds
```

```
{'t1': 10.043999136561549, 't2': 50.08724747032337, 't3': 99.91507559126703, 'N_al
```

```
INVALID PARAMETERS:
```

```
gr_chb out of bounds
```

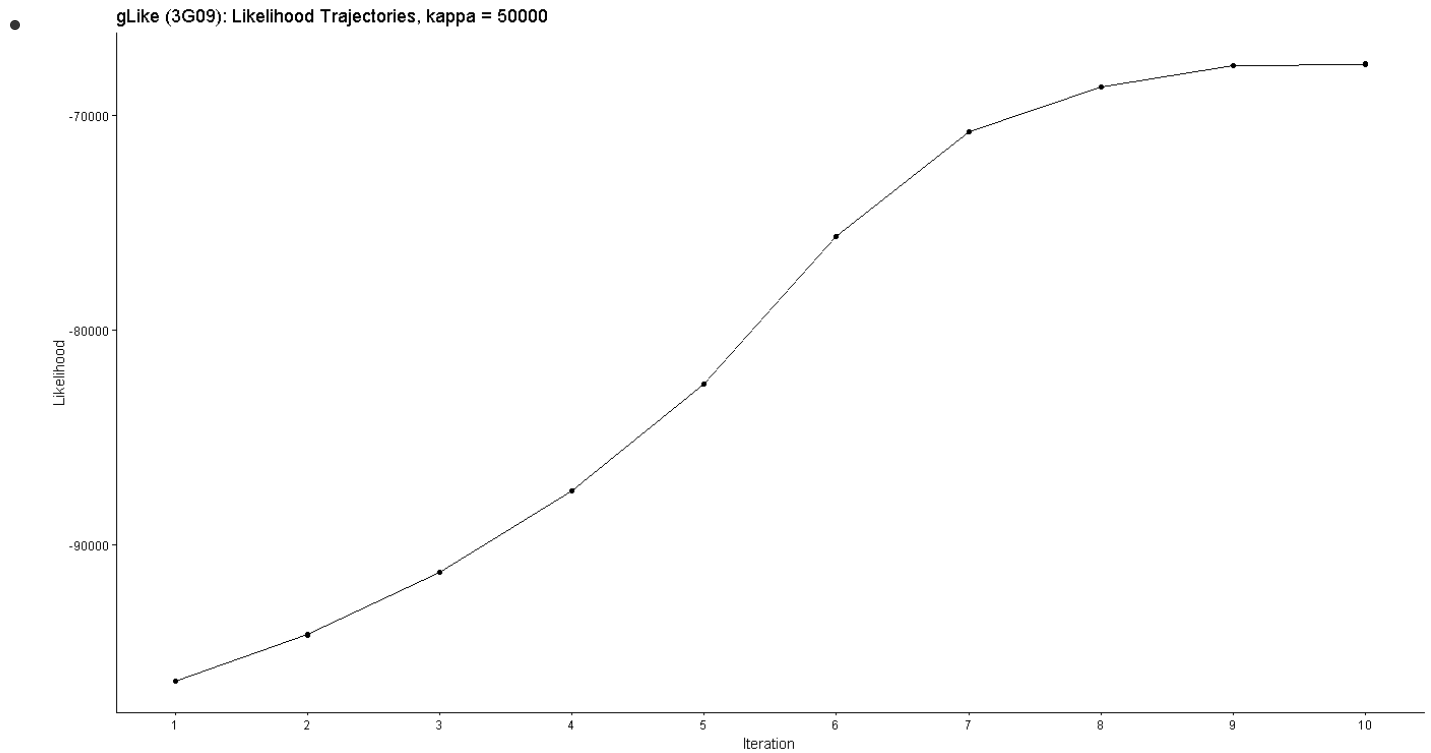
```
{'t1': 10.019683935328361, 't2': 50.00918798044388, 't3': 100.03156250358187, 'N_
```

- Example 2, migration with hard boundaries [0, 0]:

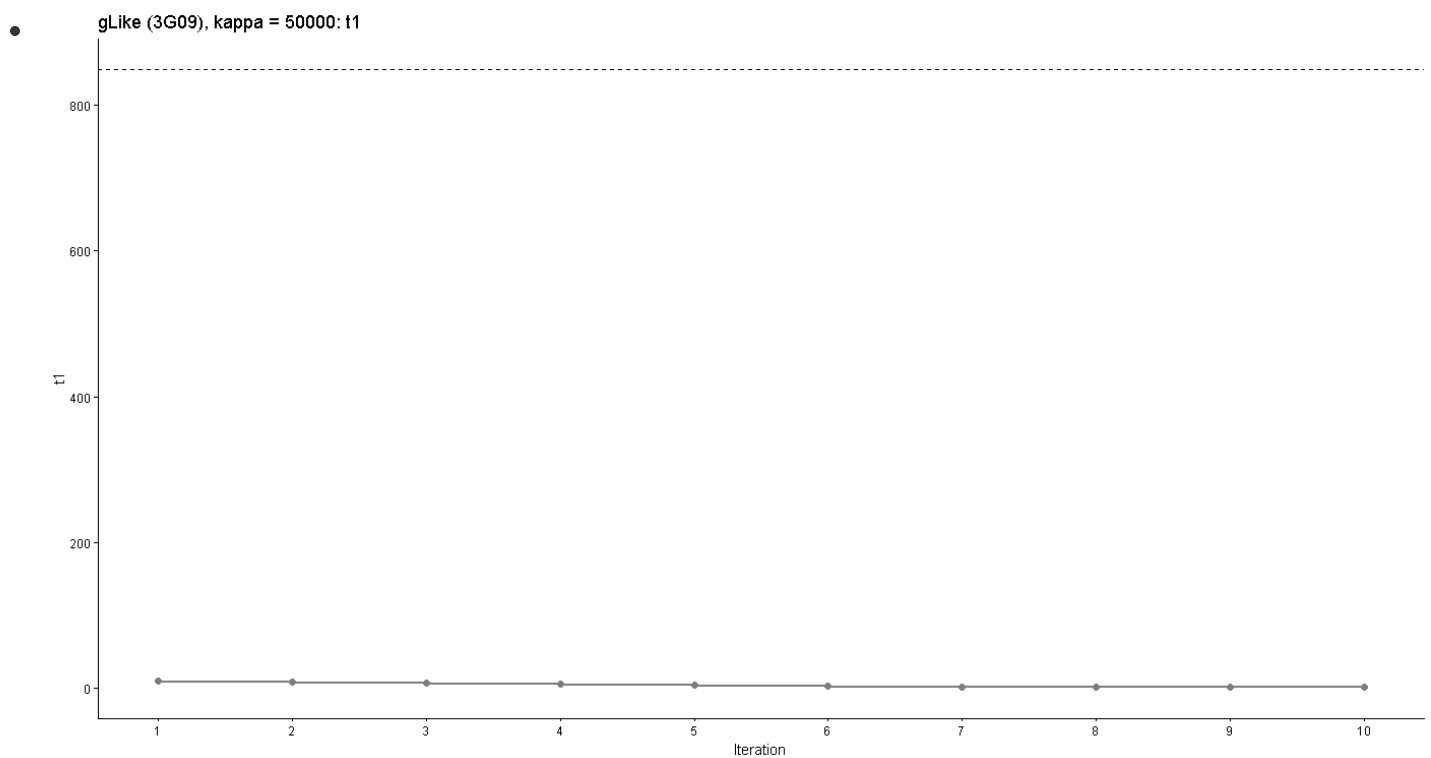
[illegible]

gLike optimization across iterations, kappa = 50,000

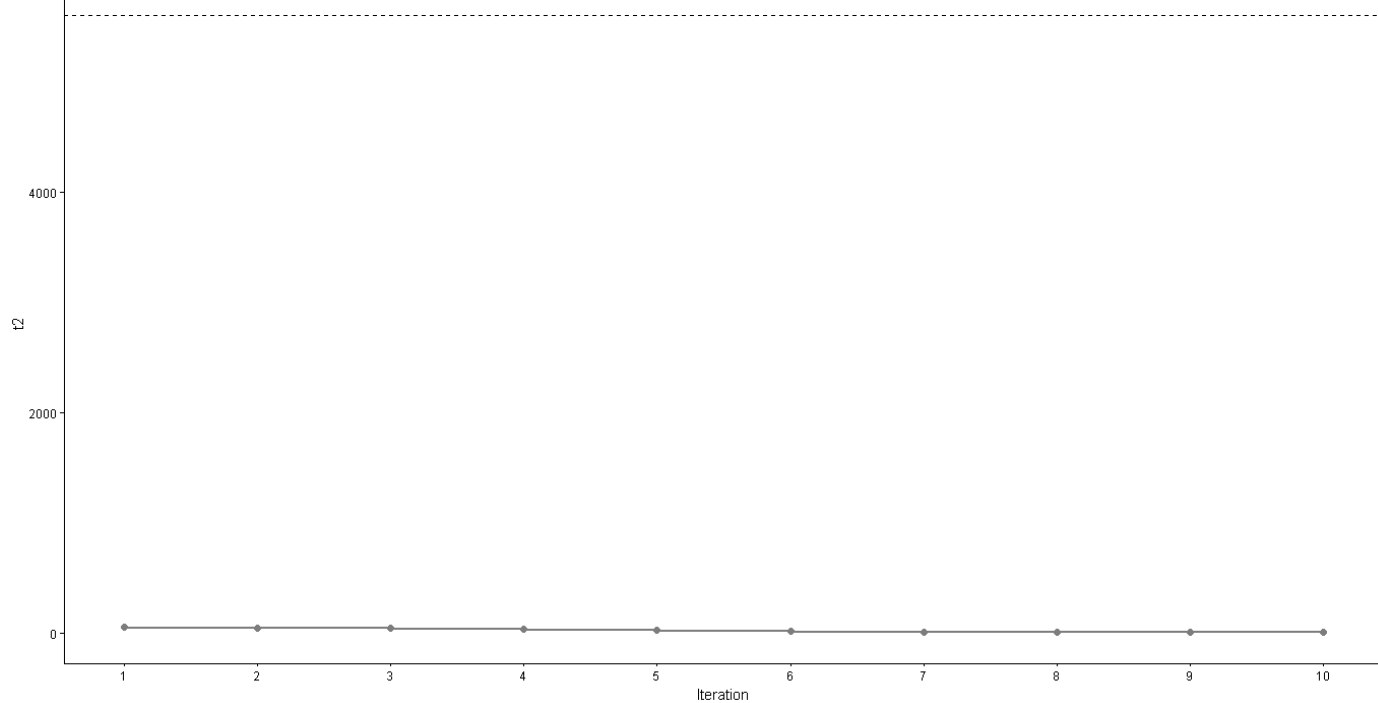
- **Note:** This is actually 10 replicates, but they are all clustered tightly together (same seed for simulation and iterator, so not unexpected)
- Note: This is the trajectory of the log likelihood of 10 replicate simulation + inferences using gLike for the 3-pop ooa (3G09) scenario.



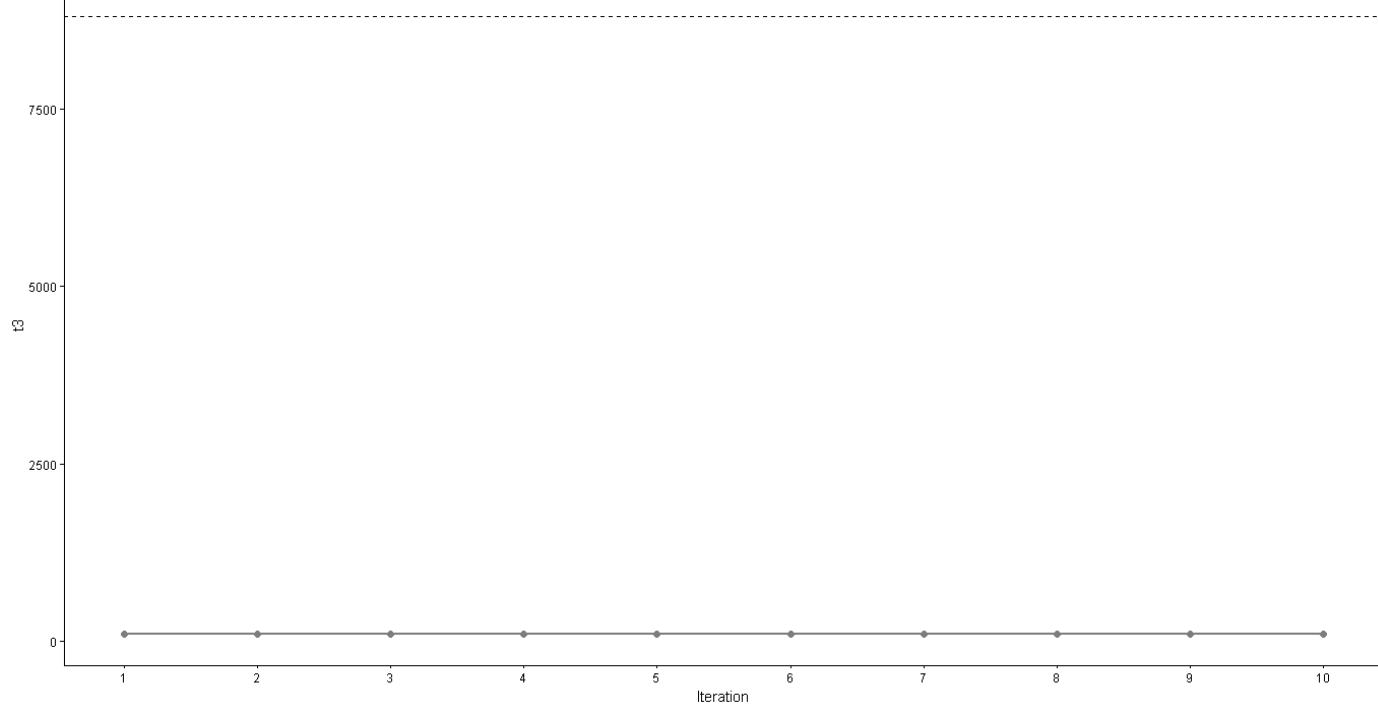
- **Note:** The dashed line indicates the true underlying value of the parameter used in the simulation.



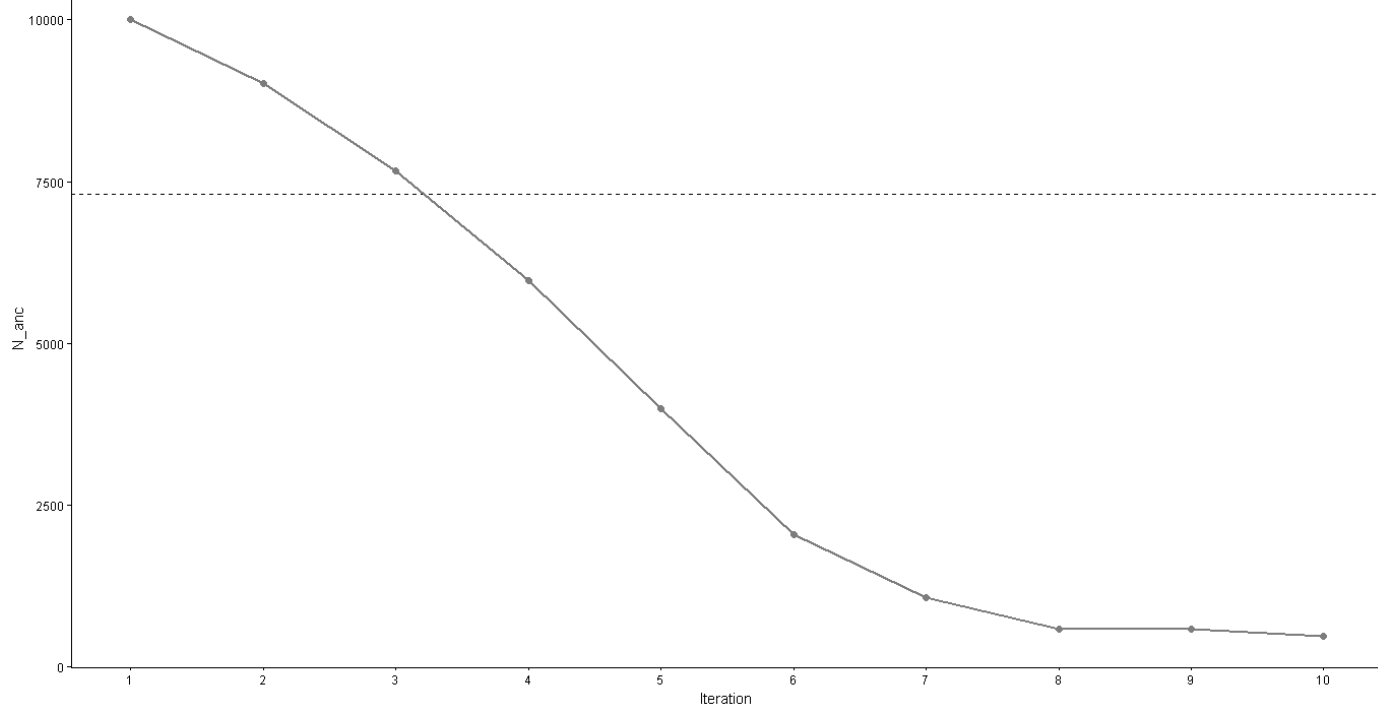
gLike (3G09), kappa = 50000: t2



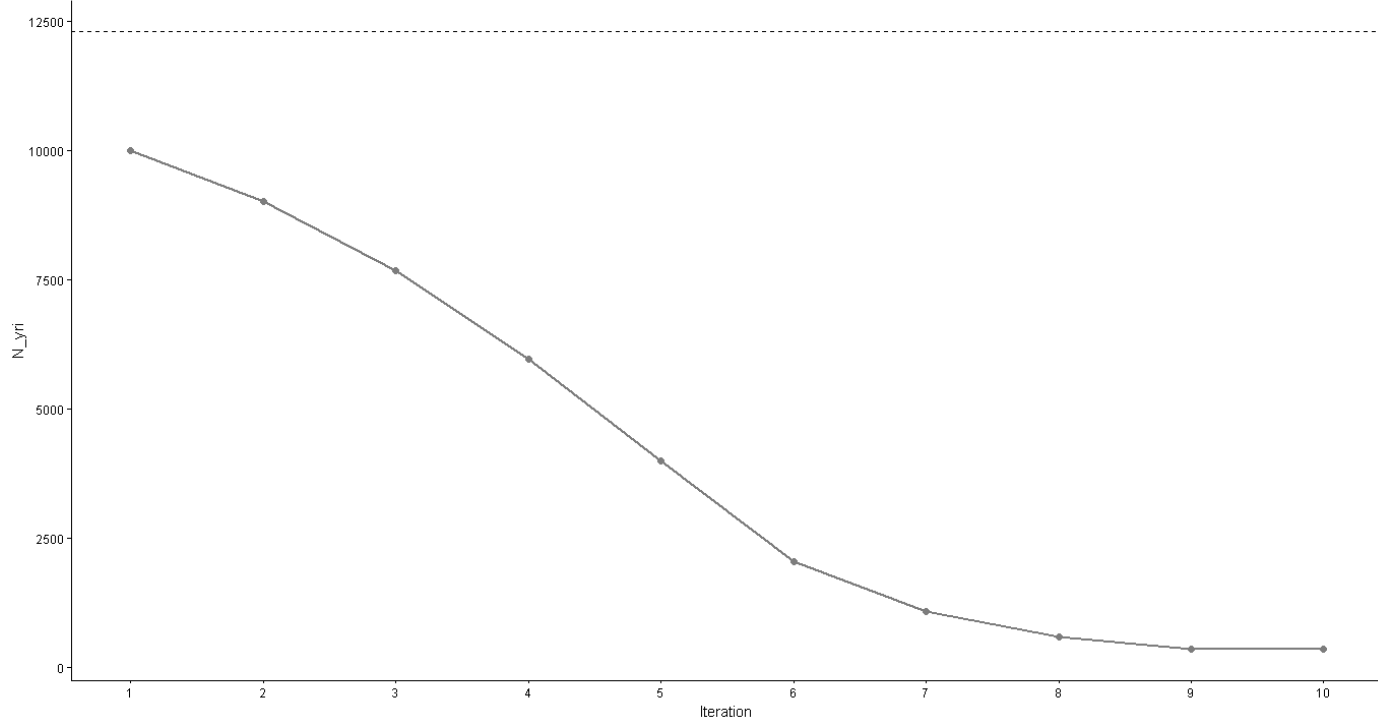
gLike (3G09), kappa = 50000: t3



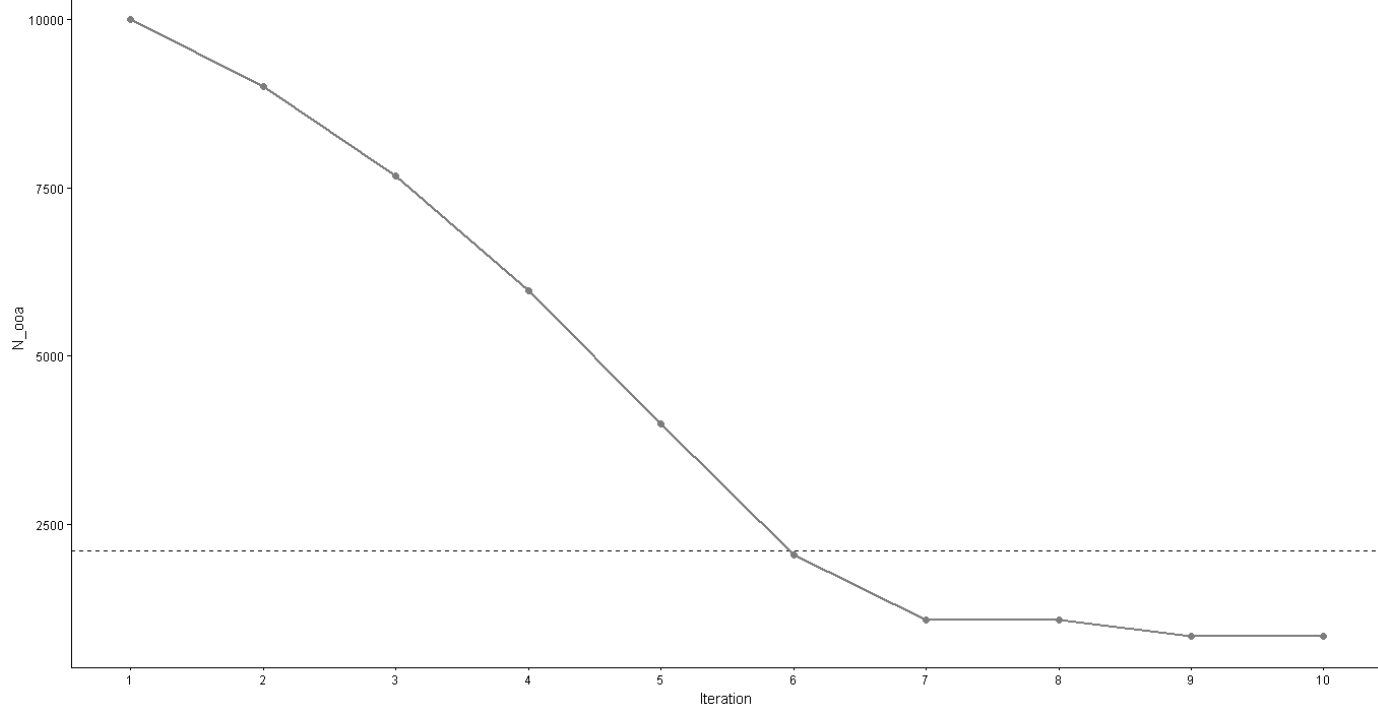
gLike (3G09), kappa = 50000: N_anc



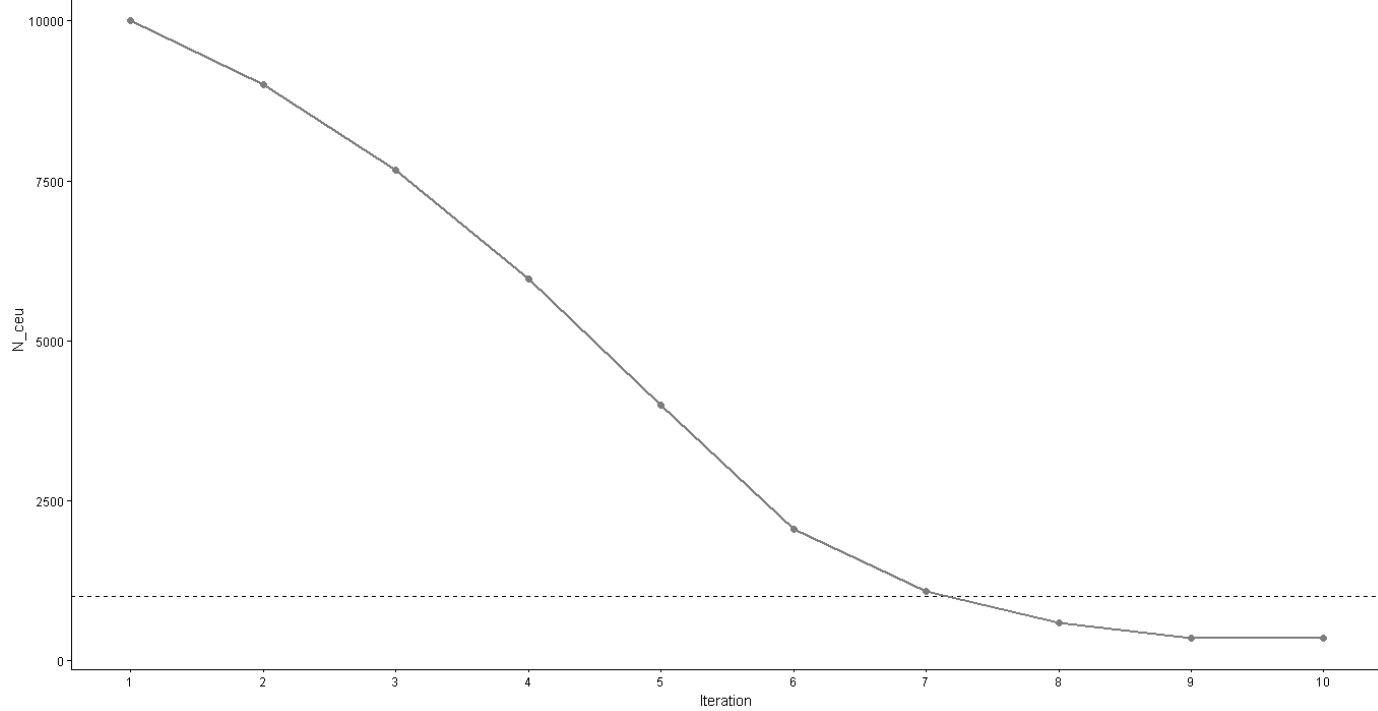
gLike (3G09), kappa = 50000: N_yri

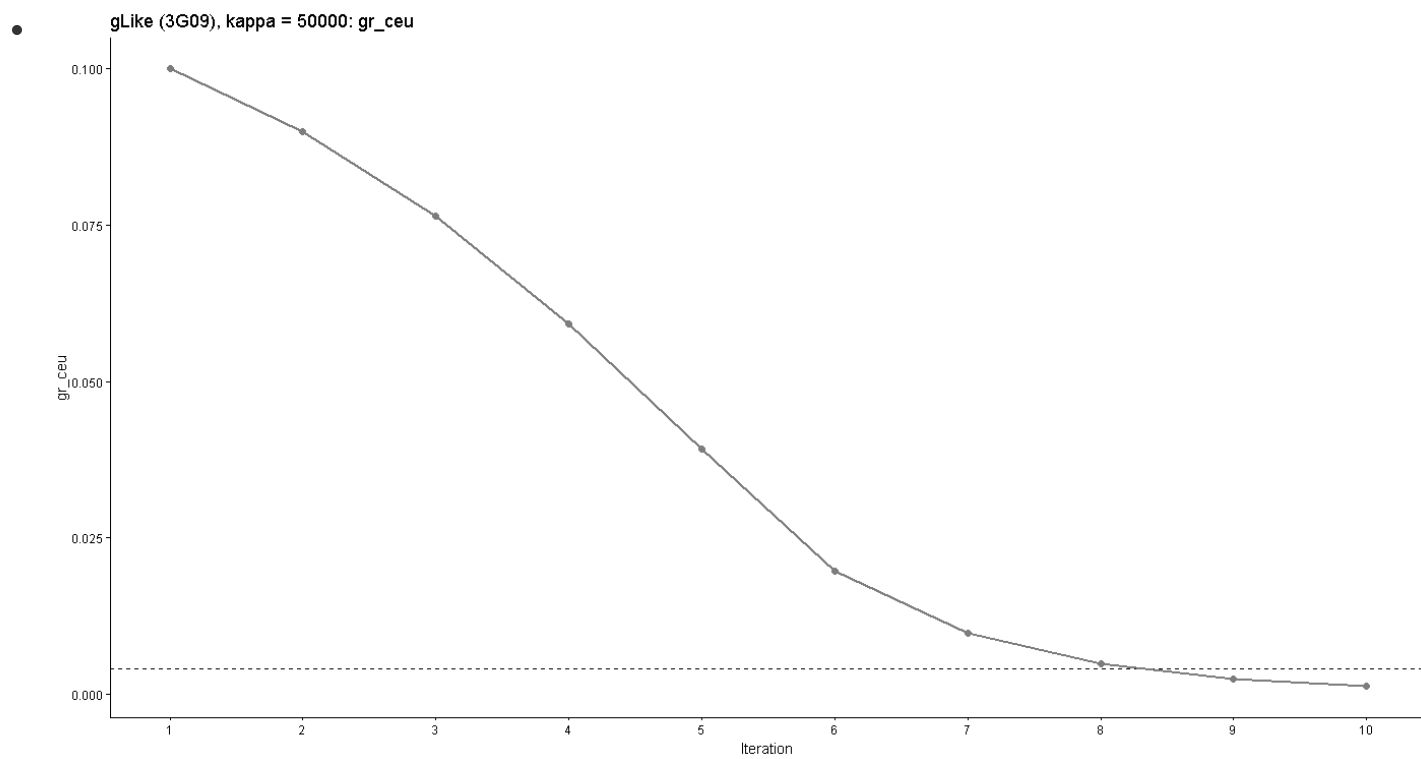
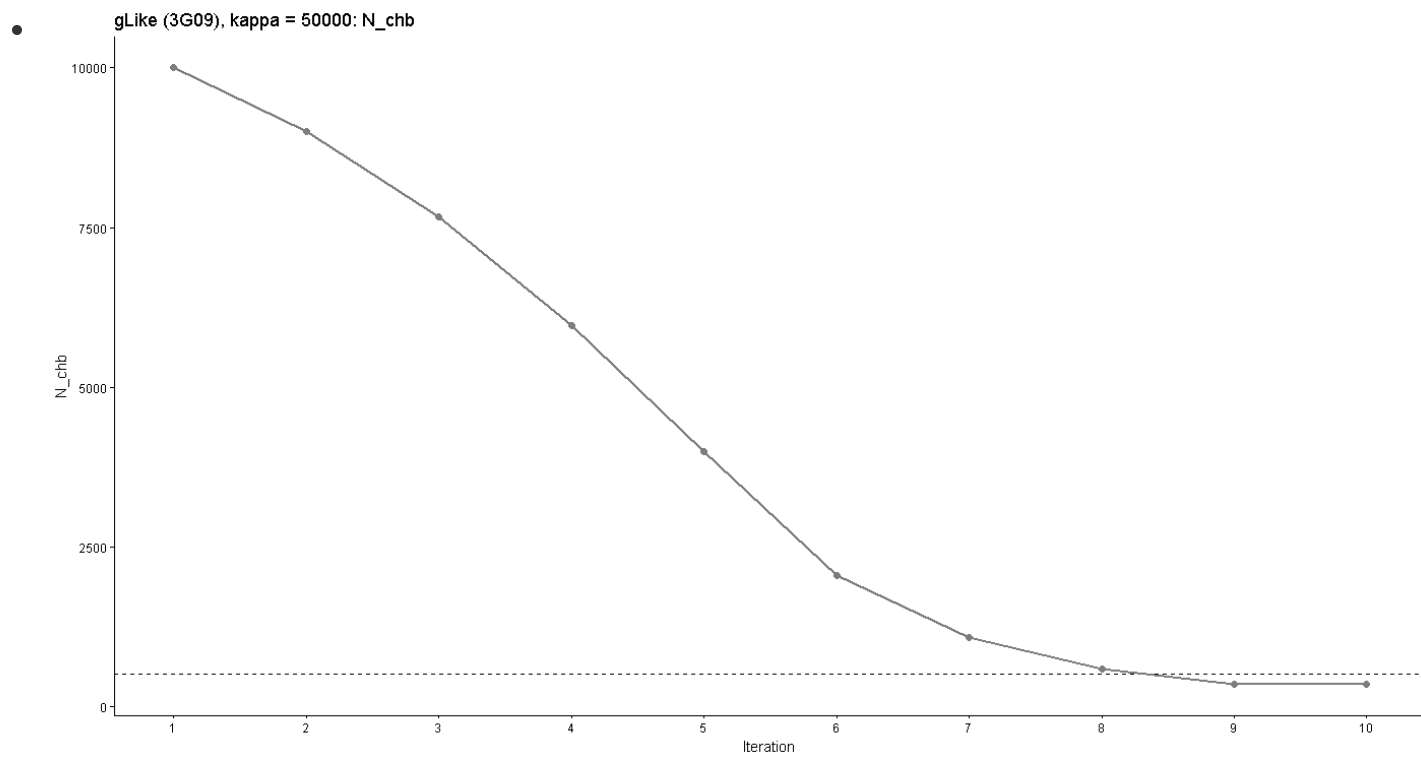


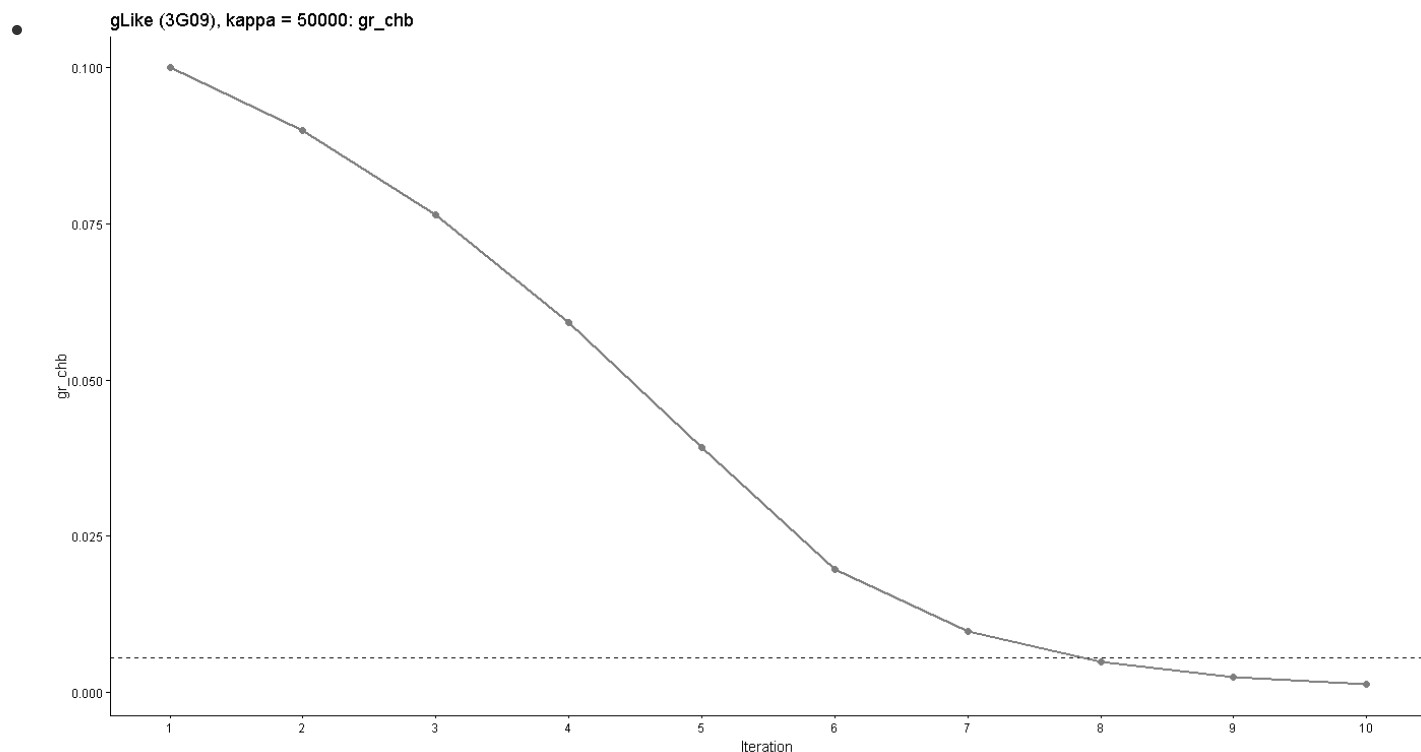
gLike (3G09), kappa = 50000: N_ooa



gLike (3G09), kappa = 50000: N_ceu

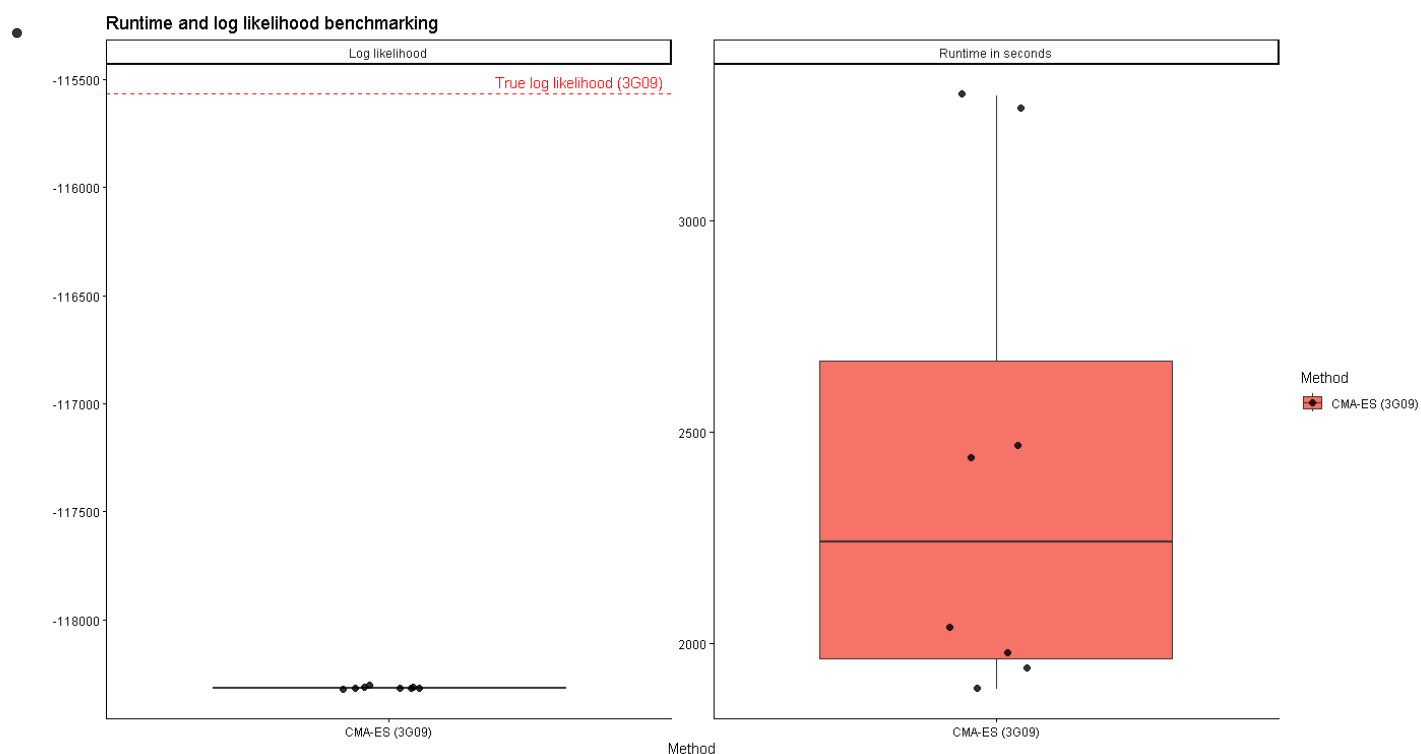




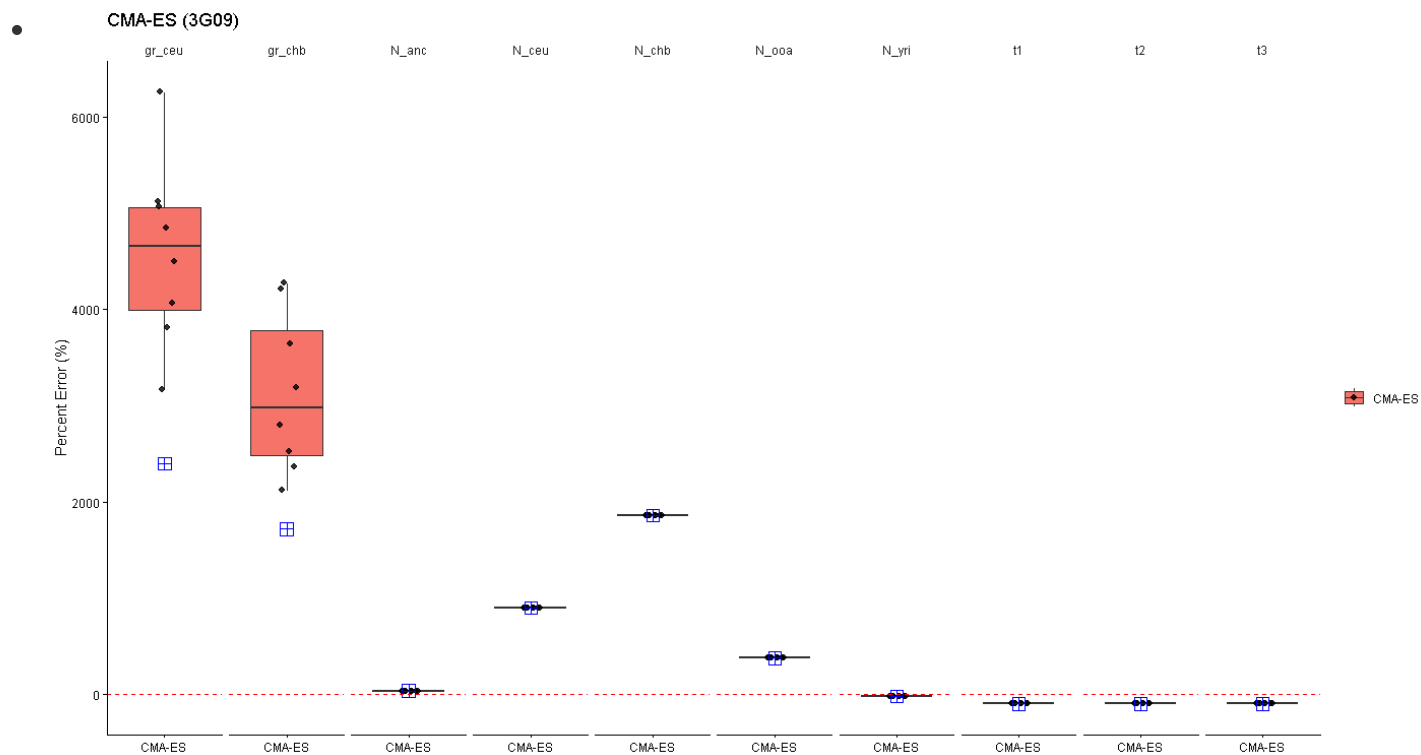


CMA-ES with kappa = 50,000

- Note:** CMA-ES doesn't have traditional "epoch" structure -- instead there are a number of generations until a tolerance condition is met. Inside the generations, I'm not sure how to print out intermediate model states, so I was unable to make the same "iterative" plots as with the gLike optimizer.



- Note:** The blue square indicates the initial value (arbitrarily chosen)



TODO

- Internal scaling
- CMA-ES
 - Initial condition not moving
 - Overall accuracy
 - 980% success rate
 - Recode soft boundaries using if conditions
- Kappa tests (10k vs 20k vs 30k etc.)